

Lean Deployment SOP v1.0

1. Purpose

- Provide a controlled and repeatable deployment process.
- Reduce production risk and prevent downtime.

2. Scope

- Applies to backend (FastAPI), frontend (React), MariaDB, Nginx, and Gunicorn deployments.

3. Roles & Permissions

- Only authorized engineer can deploy to production.
- Deployment requires PR approval.
- No direct changes on production server.

4. Environment Structure

- Dev → Staging → Production.
- Separate database and environment variables for each.
- Production secrets never shared with staging.

5. Pre-Deployment Checklist

- All tests passing.
- PR approved and merged.
- Database migration reviewed.
- Backup created (mysqldump).
- Version tag created (e.g., v1.1.0).

6. Deployment Steps

- Pull latest main branch on server.
- Activate virtual environment.
- Install dependencies if needed.
- Run migrations.
- Restart tta service (systemctl restart tta).
- Check logs and run smoke tests.

7. Rollback Procedure

- Revert to previous Git tag.
- Restore database backup if schema affected.
- Restart service and validate logs.

8. Post-Deployment Monitoring

- Monitor logs using journalctl.
- Check Nginx status.
- Verify API response and frontend loading.
- Monitor memory and disk usage.

9. Backup Policy

- Daily automated mysqldump backup via cron.
- Store backups securely and verify periodically.

10. Incident Response

- P0: Application down – immediate rollback.
- P1: Core feature broken – hotfix deployment.
- P2: Minor issue – schedule patch release.